

BHU-NETRA: Satellite Ground Segment & Mission Control Platform

Complete Enterprise Architecture & Reference Implementation

Executive Overview & Solution Summary

As Enterprise Architect for **BHU-NETRA** (national satellite ground-segment platform managing 40+ Earth Observation and Communication satellites), a complete, end-to-end architectural design and production-ready reference codebase has been created in the workspace directory .

 Assignment Task Matrix & Artifact Map

Assignment Item	Document & Code Location	Key Deliverables Summary
a) Functionality List		TT&C tracking, seamless mid-pass handover, real-time telemetry calibration, 3-stage cryptographic command enclave, high-throughput payload orthorectification, constraint-based pass planning, weather preemption.
b) Micro-services		16 decoupled microservices across 4 domains (OMPD, TFOD, PGPD, DGPD), inter-service protocol matrix (gRPC, Kafka, VITA 49, REST), domain event schemas.
c) Architecture Design		Dual Architecture Comparison: Stack A (Open Source / Indigenous: Orekit, PostGIS, MinIO, TimescaleDB, Keycloak) vs. Stack B (Enterprise Proprietary: Ansys STK, Oracle Spatial, IBM MQ, ESRI ArcGIS).
d) API Details		REST OpenAPI 3.0 catalog, gRPC mTLS schemas, WebSocket real-time telemetry stream specs, HTTP payloads for scheduling, commands, handover, STAC search, and orders.
e) Hardware Details		11m/7.5m S/X/Ka Cassegrain tracking antennas, USRP X410 SDRs, 16x Node NVIDIA H100 HPC GPU orthorectification cluster, 200TB NVMe + 5PB Warm SAN + 18PB LTO-9 Tape, Thales HSM Level 3 enclave, Advenica Data Diodes.
f) Data Flow Map & DB		8-Stage Satellite Pass Journey Data Flow Diagram , formal JSON/XML schemas, PostgreSQL/PostGIS DDL (<code>init_postgres_postgis.sql</code>), TimescaleDB SQL (<code>telemetry_timescale.sql</code>), Redis cache topology, MinIO S3 policy.
Code Implementation & E2E Simulator		4 runnable microservices (Mission Planner, Command Enclave, SDR Gateway, Dissemination Portal) and a zero-dependency E2E Test Runner simulating the complete pass journey.
Git Repository & GitHub Tooling		Local git repository initialized with 2 commits, plus automated GitHub push helper script.

 End-to-End Satellite Pass Journey Data Flow Diagram





Dual Architecture Comparison: Open Source vs. Enterprise Proprietary

NOTE

Architectural Recommendation: For national strategic security and air-gapped immunity from foreign export restrictions or vendor lock-in, **Architecture Stack A (Indigenous Open-Source)** is selected as the primary core for Mission Control and C2 operations.



Empirical Verification & Simulation Test Run

The simulation test runner (`python scripts/test_all_services.py`) was executed in the workspace and verified all 8 stages of a satellite pass's journey with **100% pass status**:

text

```
=====
      BHU-NETRA: SATELLITE PASS JOURNEY END-TO-END SIMULATION SUITE
=====

[STAGE 1] Querying Mission Planner for Satellite Pass Schedule...
-> Schedule Generated: SCH-20261009-78C5 (Total Passes: 3)
-> Pass ID: PASS-NETRA-E-SHAD-FBF1 | Sat: NETRA-EO-01 | GS: GS-SHADNAGAR-01

[STAGE 2 & 3] Ingesting Real-Time Telemetry Stream & Anomaly Check...
-> Event ID: TEL-EVT-B8029C67 | CADU Frame: #97998
-> Bus Voltage: 27.81V | Battery SoC: 93.4%
-> Anomaly Status: NOMINAL

[STAGE 4] Executing 3-Stage Air-Gapped Command Authorization Workflow...
-> Stage 1 (Operator): Command Created -> ID: CMD-AUTH-BC666E [Status:
PENDING_FDO_APPROVAL]
-> Stage 2 (FDO): Approved Orbit & Power Window [Status: PENDING_MD_APPROVAL]
-> Stage 3 (Mission Director HSM): Cryptographic Signature Generated!
-> CCSDS Telecommand Frame: 1ACFFC1D000577412001124584C814C652D1E4A5BCH_CRC_OK
-> Final State: FULLY_AUTHORIZED_FOR_UPLINK (Ready for Uplink)

[STAGE 5] Executing Mid-Pass Ground Station Control Handover...
-> Handover ID: HO-20261009-6993
-> Transfer: GS-SHADNAGAR-01 -> GS-PORTBLAIR-02
-> Link State: PHASE_LOCK_SYNCHRONIZED_SUCCESS | Telemetry Continuity: ZERO_FRAME_LOSS

[STAGE 6 - 8] Searching STAC Imagery Catalog & Ordering Data Dissemination...
-> STAC Items Found: 2
-> Selected STAC Item: NETRA_EO01_20261010_L3_AGRICOG_091 (L3 COG)
-> Delivery Envelope ID: DEL-20261009-AAE6
-> Recipient: MINISTRY_OF_AGRICULTURE via NICNET_DIRECT_GOV_BACKBONE
-> ABAC Policy Applied: Classification Mask = True
-> Storage URI: s3://bhu-netra-level3-
products/2026/10/10/NETRA_EO01_20261010_L3_AGRICOG_091.tif

=====
      SUCCESS: ALL 8 STAGES OF SATELLITE PASS JOURNEY SIMULATED SUCCESSFULLY!
=====
```

GitHub Repository Push Instructions

The workspace folder has been initialized as a local Git repository with all documentation, schemas, SQL DDL, microservices, and tests committed.

To push the codebase directly to your GitHub account, run:

```
bash
cd c:\Users\lenovo\Downloads\HyperVerge\bhu-netra
python scripts/init_git_repo.py https://github.com/YOUR_USERNAME/BHU-NETRA.git
[YOUR_GITHUB_TOKEN]
```